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2a.

```
shashanth@DESKTOP-B43QA: ~$ nano program1.c
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];
    while (1) {
        printf("myshell> ");
        if (fgets(input, sizeof(input), stdin) == NULL) break;
        input[strcspn(input, "\n")] = '\0';
        printf("You entered: %s\n", input);
    }
    return 0;
}
```

```
shashanth@DESKTOP-B43QA29:~$ mkdir practical2
shashanth@DESKTOP-B43QA29:~$ cd practical2
shashanth@DESKTOP-B43QA29:~/practical2$ nano program1.c
shashanth@DESKTOP-B43QA29:~/practical2$ nano program1.c
shashanth@DESKTOP-B43QA29:~/practical2$ gcc program1.c -o program1
shashanth@DESKTOP-B43QA29:~/practical2$ ./program1
myshell> hello
You entered: hello
myshell> ls
You entered: ls
myshell> pwd
You entered: pwd
myshell> |
```

2b

```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];
    while (1) {
        printf("myshell> ");
        if (fgets(input, sizeof(input), stdin) == NULL) break;
        input[strcspn(input, "\n")] = '\0';
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }
        printf("Command entered: %s\n", input);
    }
    return 0;
}
```

```

shashanth@DESKTOP-B43QA29:~/practical2$ nano program2.c
shashanth@DESKTOP-B43QA29:~/practical2$ nano program2.c
shashanth@DESKTOP-B43QA29:~/practical2$ gcc program2.c -o program2
shashanth@DESKTOP-B43QA29:~/practical2$ ./program2
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> exit
Exiting shell...
shashanth@DESKTOP-B43QA29:~/practical2$

```

2c

```

GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main()
{
    char input[100];

    while (1)
    {
        // Display Prompt
        printf("myshell> ");

        // Read User Input
        fgets(input, sizeof(input), stdin);

        // Remove newline
        input[strcspn(input, "\n")] = '\0';

        // Check exit condition
        if (strcmp(input, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }

        // Process Input
        printf("Command entered: %s\n", input);
    }

    return 0;
}

```

```

shashanth@DESKTOP-B43QA29:~/practical2$ nano program3.c
shashanth@DESKTOP-B43QA29:~/practical2$ nano program3.c
shashanth@DESKTOP-B43QA29:~/practical2$ gcc program3.c -o program3
shashanth@DESKTOP-B43QA29:~/practical2$ ./program3
myshell> hello
Command entered: hello
myshell> pwd
Command entered: pwd
myshell> exit
Exiting shell...
shashanth@DESKTOP-B43QA29:~/practical2$ |

```

2d

```
shashanth@DESKTOP-B43QA: X + v
GNU nano 8.7.1 program4.c *
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
int main() {
    struct termios oldt, newt;
    char ch;
    char buffer[100];
    int index = 0;
    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);
    printf("myshell> ");
    while (1) {
        ch = getchar();
        if (ch == '\n') {
            buffer[index] = '\0';
            printf("\nCommand: %s\n", buffer);
            if (index == 4 && buffer[0] == 'e' && buffer[1] == 'x' && buffer[2] == 'i' && buffer[3] == 't') {
                break;
            }
            index = 0;
            printf("myshell> ");
        } else if (ch == 127 || ch == 8) {
            if (index > 0) {
                index--;
                printf("\b \b");
                fflush(stdout);
            }
        } else if (index < 99) {
            buffer[index++] = ch;
            putchar(ch);
            fflush(stdout);
        }
    }
    tcsetattr(STDIN_FILENO, TCSANOW, &oldt);
    return 0;
}
```

2e

```
shashanth@DESKTOP-B43QA: X + v
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>
int main() {
    char buffer[100];
    while (1) {
        printf("myshell> ");
        if (fgets(buffer, sizeof(buffer), stdin) == NULL)
            break;
        buffer[strcspn(buffer, "\n")] = '\0';
        if (strcmp(buffer, "exit") == 0) {
            printf("Shell terminated.\n");
            break;
        }
        if (strcmp(buffer, "pwd") == 0) {
            printf("Current directory command received.\n");
        } else if (strcmp(buffer, "ls") == 0) {
            printf("List files command received.\n");
        } else {
            printf("Command received: %s\n", buffer);
        }
    }
    return 0;
}
```

```
shashanth@DESKTOP-B43QA29:~/practical2$ nano program5.c
shashanth@DESKTOP-B43QA29:~/practical2$ gcc program5.c -o program5
shashanth@DESKTOP-B43QA29:~/practical2$ ./program5
myshell> hello
Command received: hello
myshell> ls
List files command received.
myshell> pwd
Current directory command received.
myshell> test123
Command received: test123
myshell> exit
Shell terminated.
shashanth@DESKTOP-B43QA29:~/practical2$ |
```